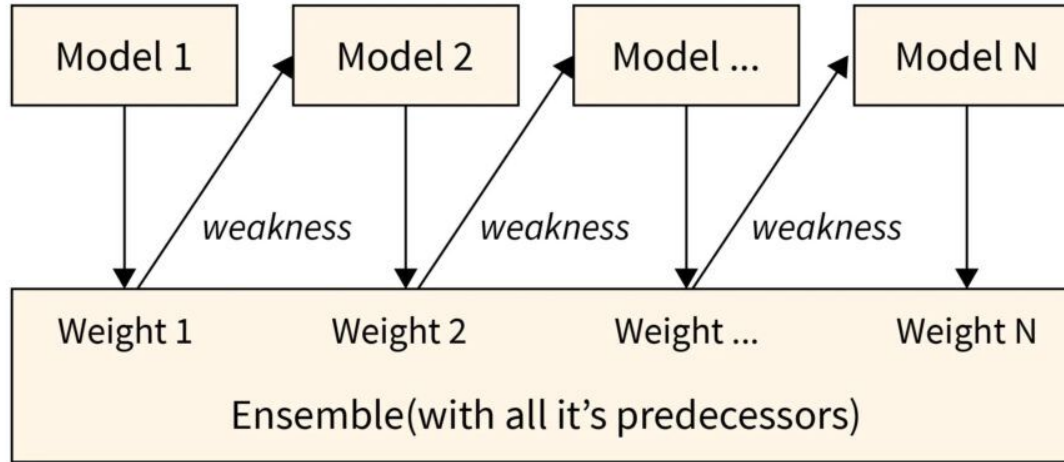


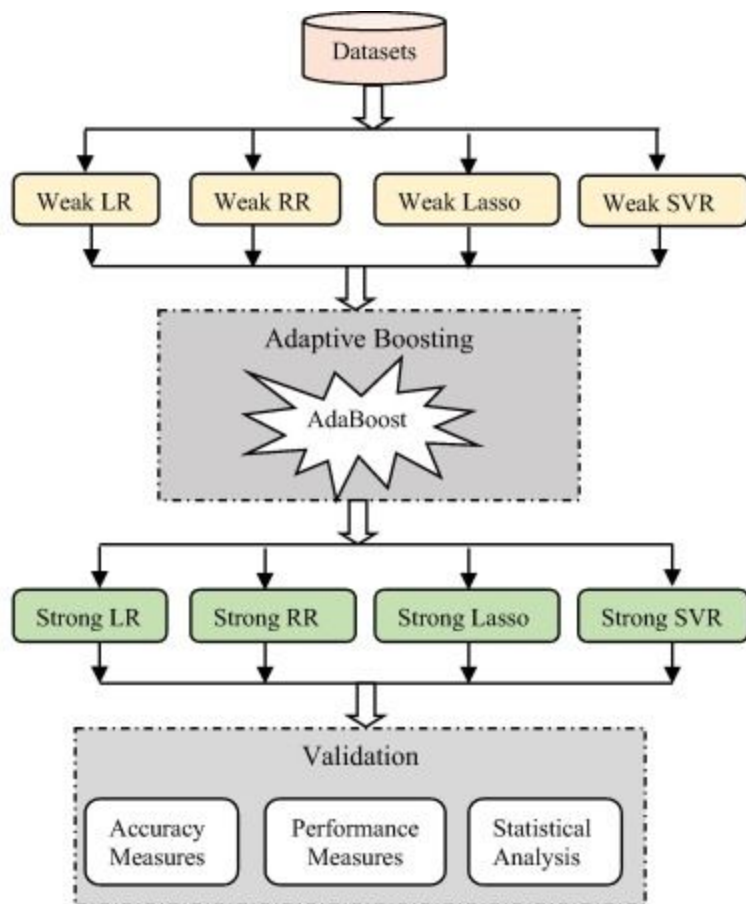
Boosting Algorithm

1. Ada boost

AdaBoost (Adaptive Boosting) is a supervised machine learning ensemble technique that combines multiple **"weak learners"** to create a single **"strong learner"**. It works sequentially, meaning each subsequent model aims to correct the errors of its predecessor.

Model 1,2,...,N are individual models (eg. decision tree)





How Ada Boos algor

1. **Train Weak Learner:** A simple model, most commonly a **decision stump** (a decision tree with only one split), is trained on the data.
2. **Calculate Error & "Amount of Say":** The algorithm calculates the weighted error rate of the stump. Based on this error, it assigns a weight (called **Alpha**) to that model; more accurate models get a higher "amount of say" in the final decision.
3. **Update Sample Weights:**
 - **Misclassified samples** receive higher weights so they are prioritized by the next learner.
 - **Correctly classified samples** receive lower weights.
4. **Repeat:** This process iterates for a specified number of learners (estimators) or until the training data is perfectly predicted.
5. **Final Prediction:** To make a prediction, AdaBoost performs a **weighted majority vote** across all weak learners. The learners with higher Alpha scores have more influence on the final result.

Characteristics of Ada Boost

- **Base Learners:** While any algorithm that accepts weighted training data can be used, **decision stumps** are the standard default.
- **Bias vs. Variance:** AdaBoost is primarily used to **reduce bias**.
- **Loss Function:** It typically minimizes an **exponential loss function**.
- **Hyperparameters:** Key parameters to tune include `n_estimators` (number of learners) and `learning_rate` (shrinks the contribution of each model)